

FOSSILS

Fossils are evidence of past life that has been buried and preserved in the rocks of Earth's crust. They give scientists important clues about how life has evolved, and they can also be used to date rocks, and create a timeline of the events that have shaped our world.



Over time the tissues of this plant have decayed. Only the outline remains, covered by a thin film of carbon.



An insect became stuck in resin oozing from a tree. The resin turned into amber, perfectly preserving the creature.



This fish skeleton has been fossilized in shale. Here all the pores in the original skeleton have been infilled by minerals.

Life has existed on Earth for at least 3.7 billion years. Initially, living things were small and soft-bodied, and left few obvious traces of their existence. But during the past billion years, life gradually changed. Organisms evolved hard body parts, which – given enough time – could fossilize. The significance of this change is hard to overstate. It turned the world's sedimentary rocks into a global data bank, teeming with an incredible array of fossilized species, arranged in the exact order in which they appeared. These fossils show the path evolution has followed. They also highlight mass extinctions – when enormous numbers of species have been wiped out in a relatively short space of time.

DEAD AND BURIED

Fossilization is a lottery, and only a small fraction of living organisms end up being preserved. On land, it is usually triggered by chance events – for example, when animals are overwhelmed by landslides or flash floods, or when they drown in lakes. Marine animals have a much better chance of being fossilized, because sediment routinely accumulates over their dead remains. Fine sediment can preserve soft bodies, but the best fossils are left by animals with hard body parts, such as shells or bones. After burial, the remains are slowly infiltrated by dissolved

minerals, which literally turns them to stone. Once formed, many fossils are destroyed by heat, pressure, or geological movements while still deep underground. But if a fossil survives all this, uplift may eventually bring it back to the surface, where erosion can release it from its parent rock (see panel, below). There, it has to be found before it eventually breaks apart.

These body fossils can be breathtaking objects, particularly when they are complete skeletons several metres long. However, they are not the only kind of fossilized remains. Rocks can also yield trace fossils, which are fossilized footprints, burrows, or other signs of animal activity. Trace fossils provide indirect but fascinating evidence of how animals lived: for example, dinosaur footprints can show how fast they moved, how they interacted in herds, and even how they put on weight as they grew.

Much further back in time, rocks sometimes contain chemical fossils – ancient carbon-based compounds that have been produced by biological processes. Although unspectacular, these chemical smudges are key evidence in the hunt for Earth's earliest living things.

SUDDEN DEATH >

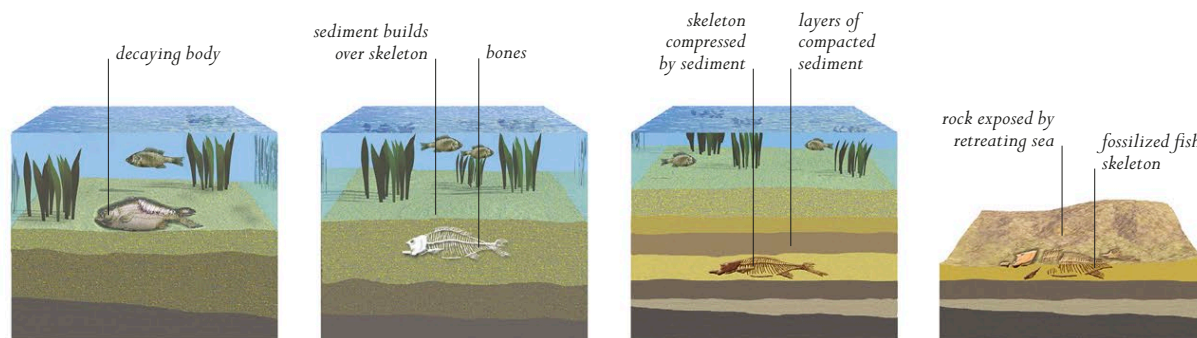
Trilobites from the Late Ordovician Period are fossilized together, suggesting the sudden burial of these animals by sediment.

DEBATE

INDEX FOSSILS

The geological timescale was largely established using fossils. Species which lived across a wide geographical range, but only existed for a short time, are known as index fossils. They can be used to identify particular strata and link them from place to place – the presence of the same index fossil in different places shows that the strata were laid down at the same time. Index fossils therefore help geologists to date rocks and build a relative time sequence. Mesozoic ammonites (a group of extinct marine molluscs) are among the best index fossils – an ammonite zone may be as little as a million years.

HOW FOSSILS FORM



A dead fish lies on the seabed, where its flesh may rot or be eaten. To be preserved, it must be buried rapidly. As the mud turns into shale, this change will compress and flatten the body.

The fish skeleton has been covered by sediment. To become fossilized, the bones must undergo a chemical change, called permineralization, where the pores in the bone are replaced by minerals.

More sediment has accumulated on the seabed and compressed the lower layers. The sediment is compressed and the fossil may become flattened, distorted, or destroyed.

Millions of years later the seabed sediment, now turned to rock, has been exposed by retreating seas. Weathering has further denuded the rock around the fossilized fish skeleton.